Práctica Tres: Sistema muscoesqueletico

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

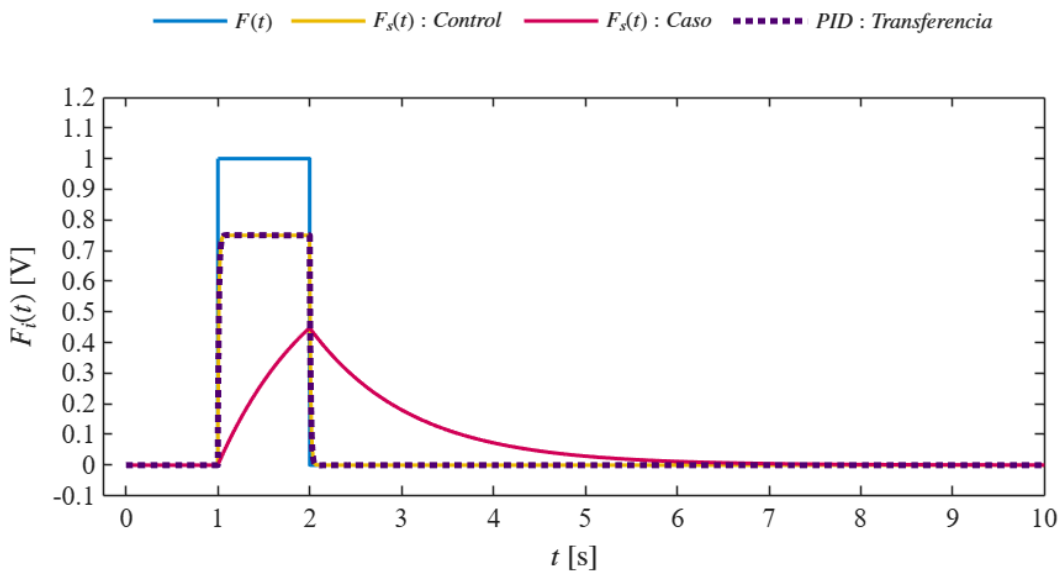
Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('MPSFP3NFGA.slx')
parameters.StopTime = '10';
parameters.Solver = 'ode15s';
parameters.MaxStep ='1e-3';
```

Respuesta al escalón

```
X = sim('MPSFP3NFGA.slx',parameters);
plotsignals(X.t,X.F,X.Fsx,X.FsY,X.PID, 'Simulink')
```



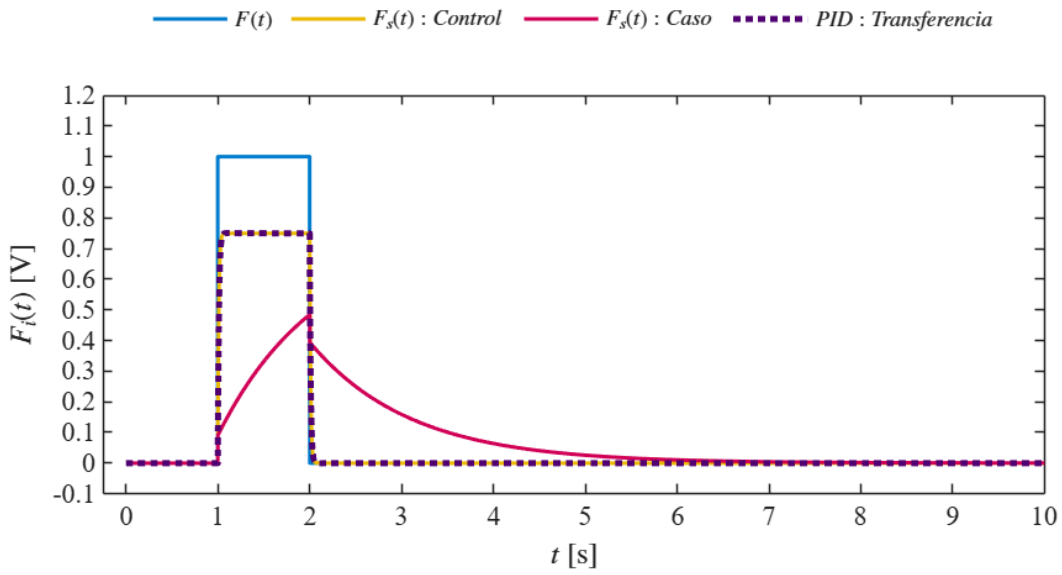
Modelo de Edos

```
dt = 1e-3;
t = (0:dt:10)';
F = zeros(length(t),1); F(1/dt:2/dt) = 1;
PID = interp1(X.t, X.PID, t, 'previous');

par_control = [0.25,10e-6,100e-6,100];
par_caso = [0.25,10e-6,100e-6,10e3];

Fs_control = musculoesqueletico(par_control,F,t);
Fs_caso =  musculoesqueletico(par_caso,F,t);

plotsignals(t,F,Fs_control,Fs_caso,PID,'EDOsys')
```




```
function Fs = musculoesqueletico(par,F,t)
    alpha = par(1); Cs = par(2); Cp = par(3); R = par (4); dt =  t(2);
    n = round(t(end)/dt);
    x = zeros(length(t),1); x(1) = 0;
    Fs = zeros(length(t),1); Fs(1) = 0;
    for i = 1:n
        x(i+1) = x(i) + ((F(i)-x(i))/(R*(Cp+Cs)))*dt;
        Fs(i+1) = Cs*F(i)/(Cp+Cs) + (1-alpha-Cs/(Cp+Cs))*x(i);
    end
end
```

Funcion: Respuesta de la simulacion

```
function plotsignals(t,F,Fsx,FsY,PID,name)
set(figure(),'color','w')
set(gcf,'units','inches','position',[1,1,7,3.5])
set(gca,'fontname','times new roman','fontsize',11)
hold on; grid off; box on;
ctrl1 = [0, 125, 204]/255;
ctrl2 = [235,185,0]/255;
ctrl3 = [209,0,86]/255;
ctrl4 = [80,0,113]/255;
order = [ctrl1;ctrl2;ctrl3;ctrl4]; colororder(order)

plot(t,F,'-','Linewidth',1.5,'Color',ctrl1)
plot(t,Fsx,'-','Linewidth',1.5,'Color',ctrl2)
plot(t,FsY,'-','Linewidth',1.5,'Color',ctrl3)
plot(t,PID,':','Linewidth',2.5,'Color',ctrl4)
L = legend('$F(t)$','$F_{s}(t):Control$','$F_{s}(t):Caso$','$PID:
Transferencia');
set(L,'interpreter','latex','FontSize',9,'location','northoutside','orientati
on','horizontal','Box','Off')
```

```
xlabel('$t$ [s]','interpreter','latex')
ylabel('$F_i (t)$ [V]','interpreter','latex')
xlim([-0.25 10]); xticks(0:1:10)
ylim([-0.1,1.2]); yticks(-0.1:0.1:1.2)



exportgraphics(gcf,[name,'_matlab','.pdf'],'ContentType','vector')



end
```